

Positive Chen representations for $\log \log N \geq 24.9$

A finite Wu–BJS proof and its reproducible audit

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Abstract

The objective is positivity, not preservation of a proposed numerical coefficient at 23.36. We use the published product cutoff 10^9 , a fixed-cutoff prime–integer rectangle estimate proved below in the required range $X \geq N^{0.398}$, and a complementary–prime sieve of level $N^{2/5}$. The latter retains the original p_2 -roughness, which makes both grouped coefficients bounded by one. Its small-conductor contribution is estimated before Perron separation; its large-conductor contribution is estimated by the multiplicative large sieve. All exceptional alternatives are covered by common worst-case envelopes. The arithmetic certificate uses Arb balls; the extension from one endpoint to a half-line is proved by analytic majorants, not inferred from endpoint sampling.

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1 Statement, notation and audit scope

For even N , let

$$D_{1,2}(N) = \#\{p < N : p \text{ is prime, } \Omega(N - p) \leq 2\}.$$

We count prime factors with multiplicity. Put

$$L = \log N, \quad c = \log L, \quad N_0 = \exp(\exp(249/10)),$$

$$C_N = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad \Theta(N) = \frac{C_N N}{L^2}.$$

The elementary bound $C_N > 1/2$ suffices throughout: deleting factors from the convergent product

$$\prod_{n=2}^{\infty} (1 - n^{-2}) = \frac{1}{2}$$

increases it, and the factors indexed by $n = p - 1$ form a proper subset. We also use the elementary upper bound $C_N < 1 + \omega(N)$ only for finite exclusions: if $q_1 < \dots < q_j$ are the odd primes dividing N , then $q_i \geq i + 2$ and $\prod_i (q_i - 1)/(q_i - 2) \leq \prod_{i=1}^j (i + 1)/i = j + 1$.

Theorem 1.1. *Every even integer $N \geq N_0$ satisfies $D_{1,2}(N) > 0$.*

The proof is given in [sections 9](#) and [10](#). No assertion at $c = 23.36$ is retained. The supplied `reply.md` is used as the audit checklist, not as a mathematical source. References to BJS mean exactly [arXiv:2207.09452v6 \[1\]](#); references to Wu mean Lemmas 9.1–9.2 of [\[2\]](#). New statements needed outside their printed ranges are proved here and are not attributed to those papers.

All prime intervals are left-closed and right-open unless an inequality explicitly says otherwise. A finite sequence is a multiset: equal integer values with different source tags retain their multiplicities. Set

$$P_M(t) = \prod_{\substack{p < t \\ p \nmid M}} p, \quad P_{\text{all}}(t) = \prod_{p < t} p,$$

$$\mathcal{A} = \{N - p : p < N \text{ prime, } p \nmid N\}, \quad \mathcal{A}_a = \{b/a : b \in \mathcal{A}, a \mid b\},$$

and write $S(\mathcal{A}_a, P_M(t))$ for the number of members of $\mathcal{A}_a$ coprime to $P_M(t)$. Thus every member of $\mathcal{A}$ is odd and coprime to N .

The retuned rational parameters are

$$\kappa_1 = \frac{191}{2000}, \quad \kappa_2 = \frac{383}{4000}, \quad \rho_0 = \frac{119}{500}, \quad \sigma_2 = \frac{6047}{20000},$$

$$\sigma_1 = \frac{121}{400}, \quad \delta = \frac{29}{25}, \quad \lambda = \frac{1}{2} - 2\kappa_1 = \frac{309}{1000}.$$

$$z_* = N^{\kappa_1}, \quad y = N^{1/3}, \quad z = N^{1/8}, \quad u_0 = 10^9, \quad \varepsilon_0 = \frac{1}{10000}, \quad \eta = \frac{1452}{10^{10}}.$$

The parameters called ε, Z, z_0, H and the Johnston terminal in the earlier manuscript are not used. In particular, there is no undefined support parameter: the complementary level is exactly $N^{2/5}$.

2 External inputs, products and model masses

2.1 The precise linear-sieve input

Let F, f be the functions in BJS Theorem 6:

$$F(s) = 2e^\gamma/s, \quad f(s) = 0 \quad (0 < s \leq 2), \quad (sF(s))' = f(s-1), \quad (sf(s))' = F(s-1) \quad (s \geq 2).$$

On $s \geq 1$, $F \geq 1$ is decreasing, and $0 \leq f \leq 1$ is increasing. These properties also follow from the nonnegative decreasing functions f_n in BJS equations defining $F = 1 + \sum_{n \text{ odd}} f_n$ and $f = 1 - \sum_{n \text{ even}} f_n$. Define

$$h(s) = \begin{cases} e^{-2}, & 1 \leq s \leq 2, \\ e^{-s}, & 2 \leq s \leq 3, \\ 3e^{-s}/s, & s \geq 3, \end{cases} \quad \mathcal{F}(s) = F(s) + 106\eta e^2 h(s), \quad f_-(s) = f(s) - 107\eta e^2 h(s).$$

The constants 106, 107 are the outward integer bounds in BJS Table 1 at reciprocal tolerance 10^5 . They remain valid for our smaller tolerance, as stated and proved there.

BJS Theorem 6 uses a tag-dependent multiplicative function g_α , with $0 \leq g_\alpha(p) < 1$. Its main term is

$$X_A = \sum_{\alpha} w_{\alpha} \prod_{p|P(w)} (1 - g_{\alpha}(p)),$$

not an aggregate multiplicative density. The product hypothesis is

$$\prod_{\substack{p \in \mathbb{P} \setminus Q \\ u \leq p < w}} (1 - g_{\alpha}(p))^{-1} < (1 + \eta) \frac{\log w}{\log u} \quad \text{for every tag and every } 1 < u < w. \quad (1)$$

Here $Q = \prod_{p \in \mathbb{Q}} p$. The upper inequality requires $D \geq w$, the lower inequality requires $D \geq w^2$, and the remainder is the sum of absolute discrepancies over $d | P(w), d < QD$.

We also use the following model-mass formulation. It records explicitly the minor extension from the printed formulation with actual total mass.

Lemma 2.1 (Finite tagged model sieve). *Suppose finite nonnegative model masses x_{α} and multiplicative g_{α} satisfy (1). Suppose the actual divisibility counts have the form*

$$A_d = \sum_{\alpha} x_{\alpha} g_{\alpha}(d) + r(d).$$

Then, with $V_{\alpha}(w) = \prod_{p|P(w)} (1 - g_{\alpha}(p))$ and $s = \log D / \log w$,

$$S(A, P(w)) \leq \mathcal{F}(s) \sum_{\alpha} x_{\alpha} V_{\alpha}(w) + \sum_{\substack{d|P(w) \\ d < QD}} |r(d)| \quad (D \geq w),$$

$$S(A, P(w)) \geq f_-(s) \sum_{\alpha} x_{\alpha} V_{\alpha}(w) - \sum_{\substack{d|P(w) \\ d < QD}} |r(d)| \quad (D \geq w^2).$$

The discrepancy at $d = 1$ is included if the model total is not the actual total. There is no requirement that the weighted average of the g_{α} be multiplicative.

Proof. Use the Rosser upper and lower coefficients in the proof of BJS Theorem 6 (equivalently its Proposition 13), including its exact small-prime presieve. They depend on $P(w)$, D , Q , not on the model masses or tags; their absolute values are at most one and their support is $d < QD$. Their divisor sums bound $\mathbf{1}_{(n, P(w))=1}$ pointwise. Finite distributivity gives

$$\sum_d \lambda^\pm(d) A_d = \sum_\alpha x_\alpha \sum_d \lambda^\pm(d) g_\alpha(d) + \sum_d \lambda^\pm(d) r(d).$$

BJS bounds each inner model sum by $\mathcal{F}(s)V_\alpha(w)$ or $f_-(s)V_\alpha(w)$ under (1). Taking the absolute value only of the final discrepancy sum proves both assertions. This also proves the multiset version without identifying unequal tag densities. $\square$

2.2 The published 10^9 product cutoff

Lemma 2.2. *For every sieve used here, choose $\mathbb{Q} = \{p \in \mathbb{P} : p \leq 10^9\}$. Then (1) holds with the stated η , and*

$$\log Q < \bar{q} := 1000000050. \quad (2)$$

This includes densities $1/(p-1)$, densities obtained by setting any of them to zero, and the integer density $1/p$.

Proof. BJS Lemma 18 states, for $w > \exp 4000$ and $10^9 < u < w$,

$$\prod_{u \leq p < w} (1 - 1/(p-1))^{-1} < (1 + \eta) \log w / \log u.$$

All our sift limits satisfy $\log w \geq \kappa_1 e^{24.9} > 4000$. Deleting primes or decreasing their densities decreases this product. If $1 < u \leq 10^9$, choose $v > 10^9$ below both w and the least prime above 10^9 . The primes outside $\mathbb{Q}$ are unchanged by replacing the lower endpoint u by v . The published inequality at v has a smaller right side than that required at u . This proves the full quantifier in (1); the empty product case is immediate. Finally BJS Lemma 25 gives

$$\log Q \leq \vartheta(10^9) < 10^9 \left(1 + \frac{9 \cdot 10^{-7}}{\log 10^9}\right) < 1000000050.$$

The last strict inequality is checked with Arb in the structural certificate. Thus the unproved 10^7 product lemma is replaced, not silently assumed. $\square$

Set

$$\theta = \frac{2423257}{5000000}, \quad \Delta = \frac{81}{10^9}, \quad \theta_- = \theta - \kappa_1 \Delta, \quad H_N = \frac{\sqrt{x_1(N)}}{\log^{10} x_1(N)}, \quad x_1(N) = N/L^5. \quad (3)$$

The certificate verifies at $c = 24.9$ that

$$\theta < \frac{1}{2} - \frac{\bar{q} + 20c}{e^c}, \quad \frac{20c}{\kappa_1 e^c} < \Delta.$$

The first right side increases and the second decreases on $c \geq 24.9$. Consequently, on the entire half-line,

$$QN^\theta \leq \sqrt{N}/L^{20} < H_N, \quad \frac{20c}{\log w} \leq \Delta \quad (w \geq z_*). \quad (4)$$

Indeed $\log H_N \geq L/2 - 12.5c > L/2 - 20c$.

2.3 Prime-counting and arithmetic-progression inputs

Here are the remaining external inputs, with the exact conditions needed. BJS Lemmas 17, 24, 26, 29, 31, 34, 35 and 42–43 are used; the names of their formulae below remove any ambiguity about numbering.

1. For $x > \exp 4000$, the reciprocal-prime sum differs from $\log \log x + M$ by a quantity in $[-2.964 \cdot 10^{-6}/\log x, 1.436 \cdot 10^{-16}/\log x]$. Also $\sum_{d \leq t} \mu^2(d)/\varphi(d) \leq 1.1 \log t$ for $t \geq 10^9$.
2. The explicit prime-counting inequalities used in BJS give

$$N/L < |\mathcal{A}| < 1.00005N/L, \quad \pi(x) < x/(\log x - 3/2) \quad (5)$$

in all ranges used below (every such $\log x$ is larger than 4000). The hypotheses $\log \log x_1(N) \geq 10.4$ are much weaker than $c \geq 24.9$.

3. For $x_2(T) = T/\log^{15} T$, $T > X$, and $r < \log^{10} x_2(T)$, BJS Lemma 29 gives

$$\sum_{\substack{\chi \bmod r \\ \chi \neq \chi_0}} |\pi(T, \chi)| < v_r(X)T/\log^5 T, \quad \log \log x_2(X) \geq 10.4. \quad (6)$$

Its displayed formula is reproduced in [equation \(19\)](#) below.

4. Let k be the exceptional conductor up to $Q_1(x_1(N)) = \log^{10} x_1(N)$, setting $k = 0$ if it is absent or not coprime to N . A nonzero relevant conductor is odd and square-free. If $k < \log^\delta x_1(N)$, BJS's small-exceptional AP estimate applies to all square-free $d \leq H_N$ coprime to N . In the other case its nonexceptional estimate applies to those d with $k \nmid d$, and its relative estimate applies, with this fixed full conductor k , to

$$\pi(N; kd, N) - \pi(N; k, N)/\varphi(d), \quad d \leq H_N/k, \quad (d, kN) = 1.$$

We use the relative estimate only for k itself, not for an arbitrary additional large source tag.

In the notation $r(d) = |\mathcal{A}_d| - |\mathcal{A}|/\varphi(d)$ and $r_k(d) = |\mathcal{A}_{kd}| - |\mathcal{A}_k|/\varphi(d)$, these AP estimates imply

$$\sum |r(d)| \leq b(N)N/L^3, \quad \sum |r_k(d)| \leq b(N)N/L^3, \quad b(N) < 2.21/L^2, \quad (7)$$

over their respective ranges. For the first sum in the large case, $k \nmid d$ is imposed. The coefficient $b(N)$ is the BJS c_4 formula with a weaker zero bound, reproduced in [section A](#). Passing from π to $\mathcal{A}$ costs at most $\omega(N)$ per modulus; the c_4 term already includes this. Neither this passage nor the relative estimate requires that all prime factors of d be below a particular sift limit. This is also immediate from their proofs, which sum over all square-free moduli. All scale and zero-bound conditions are verified in [section 10](#) and in the structural certificate.

For the remaining elementary external inputs we use exactly the forms quoted and used by BJS:

- Brun–Titchmarsh: $\pi(x; q, a) < 2x/(\varphi(q) \log(x/q))$ for $(a, q) = 1$, $x > q$. For its direct uses here the moduli are at most L^{10} or H_N , and the arguments are at least $x_1(N)$, so $x > q$ holds. The smaller endpoints internal to BJS Lemma 29 are covered by that lemma, whose separate scale hypothesis is verified below.
- Rosser–Schoenfeld: $q/\varphi(q) < e^\gamma \log \log q + 2.5/\log \log q$ for $q \geq 3$. We use it only for an odd exceptional conductor whose $\log \log q > 2$.
- The full Mertens product estimate used in BJS's proof of Lemma 38 has relative error at most $1/(2 \log^2 x)$ for $x \geq 285$; using $1/\log^2 x$ permits either endpoint convention.

- The multiplicative large sieve in the proof of BJS Lemma 32: for any complex coefficients on a prefix of length M ,

$$\sum_{r \leq R} \frac{r}{\varphi(r)} \sum_{\chi \bmod r}^* \left| \sum_{n \leq M} a_n \chi(n) \right|^2 \leq (M + 3R^2) \sum_{n \leq M} |a_n|^2.$$

Here the star restricts to primitive characters. This inequality requires neither $R^2 \leq M$ nor a prime condition on the coefficients. For $r \in [R, 2R)$ it yields the multiplier $M + 12R^2$ used below.

These four inputs require no distribution assertion beyond their displayed ranges. The new rectangular and switched applications are proved below, rather than quoted as further external theorems.

For reference, the complete external-input condition ledger is below. It is a logical ledger: a row names every hypothesis used later and the place where it is discharged.

Input	Hypotheses actually used	Discharge in this proof
Wu Lemmas 9.1–9.2	$0 < \kappa_1 < \kappa_2 < \rho_0 < \sigma_2 < \sigma_1 < 1/3$ and $3\kappa_1 + \rho_0 \geq 1/2$	Exact rational checks precede theorem 3.1 ; the finite pointwise and Buchstab proof is repeated there, so Wu’s asymptotic O -term is not imported.
BJS Theorem 6	Nonnegative finite sequence, $0 \leq g_\alpha(p) < 1$, the product condition for every tag and $1 < u < w$, $D \geq w$ (upper) or $D \geq w^2$ (lower), and discrepancies for $d < QD$	Theorems 2.1 and 2.2 ; every later application states its level and remainder support. The tag masses are not replaced by an aggregate density.
BJS product and Chebyshev bounds	$w > \exp 4000$, $10^9 < u < w$, and the product Q of the presieving primes	Theorem 2.2 and equation (4) ; the cases $u \leq 10^9$ and an empty product are proved separately.
BJS Lemma 29	Endpoint $T > X$, $\log \log x_2(X) \geq 10.4$, and $r < \log^{10} x_2(T)$, with the applicable exceptional-zero bound	Equation (19) and theorems 5.1 and 8.2 ; both endpoints are kept, and the fixed cutoffs are strictly below the required powers.
BJS small-modulus AP estimate	$k < \log^\delta x_1(N)$, square-free $d \leq H_N$, $(d, N) = 1$	Equation (7) and section A ; the endpoint maximum and its half-line propagation are proved in section 10 .
BJS nonexceptional AP estimate	$k \geq \log^\delta x_1(N)$ and $k \nmid d$	The first terms of (11) ; the missing q_{j+1} makes $k \nmid am_j d$ exactly.
BJS relative AP estimate	One fixed nontrivial full conductor k , $(d, kN) = 1$, and $d \leq H_N/k$	Only the last term of (11) ; here k is the full exceptional conductor and $ad \leq H_N/k$.
Multiplicative large sieve	Primitive characters and arbitrary finite complex coefficient sequences	Theorems 5.1 and 8.2 ; the four positive terms after Cauchy–Schwarz and every dyadic interval are explicitly summed.

3 A finite signed Wu reduction

Define U_i as follows. All summation variables are primes and all displayed source products are required to be coprime to N . For brevity $S_M(a, w) = S(\mathcal{A}_a, P_M(w))$.

$$\begin{aligned} U_1 &= \sum_{z_* \leq p < N^{\kappa_2}} S_N(p, p), & U_2 &= \sum_{z_* \leq p < N^{\sigma_1}} S_N(p, z_*), & U_3 &= \sum_{z_* \leq p < N^{\sigma_2}} S_N(p, z_*), \\ U_4 &= \sum_{z_* \leq p_1 < p_2 < N^{\kappa_2}} S_N(p_1 p_2, z_*), \\ U_5 &= \sum_{z_* \leq p_1 < N^{\kappa_2} \leq p_2 < N^\lambda / p_1} S_N(p_1 p_2, z_*), \\ U_6 &= \sum_{z_* \leq p_1 < N^{\kappa_2}, N^{\rho_0} \leq p_2 < N^{\sigma_2}} S_N(p_1 p_2, p_1). \end{aligned}$$

For $i = 7, 8$, let

$$U_i = \sum_{N^{\tau_i} \leq p_1 < p_2 < \sqrt{N/p_1}} S_{Np_1}(p_1 p_2, p_2), \quad \tau_7 = \sigma_1, \quad \tau_8 = \sigma_2.$$

The remaining rows are

$$\begin{aligned} U_9 &= \sum_{z_* \leq p_1 < N^{\sigma_1} \leq p_2 < \sqrt{N/p_1}} S_{Np_1}(p_1 p_2, p_2), \\ U_{10} &= \sum_{N^{\kappa_2} \leq p_1 < N^{\sigma_2} \leq p_2 < \sqrt{N/p_1}} S_{Np_1}(p_1 p_2, N^{\sigma_1}), \\ U_{11} &= \sum_{N^{\kappa_2} \leq p_1 < p_2 < p_3 < N^{\sigma_2}} S_N(p_1 p_2 p_3, p_2), \\ U_{12} &= \sum_{\substack{N^{\kappa_2} \leq p_1 < N^{\sigma_2} \\ N^{\sigma_1} \leq p_2 < p_3 < \sqrt{N/p_1}}} S_{Np_1}(p_1 p_2 p_3, p_2), \\ U_{13} &= \sum_{z_* \leq p_1 < p_2 < p_3 < p_4 < N^{\kappa_2}} S_N(p_1 p_2 p_3 p_4, p_2), \\ U_{14} &= \sum_{z_* \leq p_1 < p_2 < p_3 < N^{\kappa_2} \leq p_4 < N^\lambda / p_3} S_N(p_1 p_2 p_3 p_4, p_2), \\ U_{15} &= \sum_{\substack{N^{\kappa_2} \leq p_1 < N^{\sigma_2} \\ N^{\sigma_2} \leq p_2 < p_3 < p_4 < N^{\sigma_1}}} S_N(p_1 p_2 p_3 p_4, p_3). \end{aligned}$$

Lemma 3.1. For $N \geq N_0$,

$$\begin{aligned} 4D_{1,2}(N) &\geq 4S(\mathcal{A}, P_N(z_*)) - U_1 - U_2 - U_3 + U_4 + U_5 \\ &\quad - 2U_7 - 2U_8 - U_9 - U_{10} - U_{13} - U_{14} - E_W, \end{aligned} \tag{8}$$

where, with $W = L/\log 2$ and $B = 2N^{11/12} + \sqrt{N} + W + 1$, $E_W \leq (4 + W^2)B$.

Proof. All of Wu's parameter inequalities hold; in particular

$$3\kappa_1 + \rho_0 = \frac{1049}{2000} > \frac{1}{2}.$$

This is Wu's Lemma 9.2 condition and is also the exact geometric inequality used below to truncate the positive double sum. First restrict to square-free source members coprime to N

and sifted at z_* . Wu's Lemma 9.1 is a pointwise inequality on these members. Here is a direct verification, using our quotient definition of $\mathcal{A}_d$. For any $0 < \kappa < \sigma < 1/3$ and a square-free, N^κ -rough source member a with at least three prime factors, let j be its number of prime factors below N^σ . Define $s_1(a)$ to count its prime divisors in $[N^\kappa, N^\sigma)$. Define $s_2(a)$ to count pairs $p_1 < p_2$ dividing a with $N^\sigma \leq p_1$, $p_2 < \sqrt{N/p_1}$, and all prime factors of $a/(p_1 p_2)$ at least p_2 . Define $s_3(a)$ by the same conditions except $N^\kappa \leq p_1 < N^\sigma \leq p_2$. Finally $s_4(a)$ counts triples $N^\kappa \leq p_1 < p_2 < p_3 < N^\sigma$ dividing a for which all prime factors of $a/(p_1 p_2 p_3)$ are at least p_2 . All selected factors are coprime to N , since a is. The four counts in the pointwise weight $1 - s_1(a)/2 - s_2(a) - s_3(a)/2 + s_4(a)/2$ are therefore

j	s_1	s_2	s_3	s_4
0	0	1	0	0
1	1	0	1	0
2	2	0	0	0
$j \geq 3$	j	0	0	$j - 2$

Indeed a sift at the second selected prime forces the first two selected factors to be the least two factors of a ; the optional third selected factor has $j - 2$ choices. The restriction $p_2 < \sqrt{N/p_1}$ is automatic for these least two factors, since $a \geq p_1 p_2 p_3 > p_1 p_2^2$ and $a < N$. Every row of the table gives weight zero. If $\Omega(a) \leq 2$, $s_4(a) = 0$ and the weight is at most one. This proves the needed pointwise inequality with no asymptotic error.

The finite Buchstab identity used throughout is

$$S(\mathcal{A}_d, P_M(v)) = S(\mathcal{A}_d, P_M(u)) - \sum_{\substack{u \leq q < v \\ q \nmid M}} S(\mathcal{A}_{dq}, P_M(q)).$$

It assigns a residual failing the larger sift to its least prime in $[u, v)$. Repeated selected primes contribute zero on the square-free source. Deleting a selected prime from P_M has no effect on such a residual. These observations explain all P_N versus P_{Np_1} uses in the following finite algebra.

For clarity, the finite algebra in Lemma 9.2 has the following three stages. Applying Lemma 9.1 with (κ_2, σ_2) and using Buchstab's identity gives

$$2D \geq 2S - U_1 - U_3 + U_4 + U_5 + U_6 - 2U_8 - U_{10} + U_{11} + U_{12} + \Delta_1.$$

Here Δ_1 is the negative sum of the three triple-prime domains

$$\begin{aligned} z_* &\leq p_1 < p_2 < p_3 < N^{\kappa_2}, & \text{sift at } p_1, \\ z_* &\leq p_1 < p_2 < N^{\kappa_2} \leq p_3 < N^\lambda/p_2, & \text{sift at } p_1, \\ N^{\kappa_2} &\leq p_1 < N^{\sigma_2} \leq p_2 < p_3 < N^{\sigma_1}, & \text{sift at } p_3. \end{aligned}$$

The first two use P_N and the last uses P_{Np_1} . The truncation of the positive double sum at N^λ/p_1 is valid because $p_1 \geq N^{\kappa_1}$ and $3\kappa_1 + \rho_0 > 1/2$, whence $N^{\rho_0} \geq N^\lambda/p_1$. The second use of Lemma 9.1, with (κ_1, σ_1) , gives

$$2D \geq 2S - U_2 - 2U_7 - U_9 + T, \quad T = \sum_{z_* \leq p_1 < p_2 < p_3 < N^{\sigma_1}} S_N(p_1 p_2 p_3, p_2).$$

Each of the three domains above is contained in the domain of T . Subtracting their sift limits by the finite Buchstab identity gives $\Delta_1 + T \geq -U_{13} - U_{14} + U_{15}$. Explicitly, in the first two domains the additional prime q obeys $p_1 < q < p_2$: relabel (p_1, q, p_2, p_3) as the four increasing primes. The new residual is sifted at q , the second prime, and in the second domain the old bound $p_3 < N^\lambda/p_2$ becomes exactly the bound defining U_{14} . In the third domain, $p_2 < q < p_3$ and the relabeling (p_1, p_2, q, p_3) gives U_{15} with sift at its third prime. The endpoints $q = p_1$ or $q = p_2$

vanish by square-freeness; the parts of T outside the three disjoint domains are nonnegative. This proves the fifteen-row inequality on the restricted source. All square-divisor error terms in Wu's printed asymptotic statement vanish on this square-free source.

Delete the four nonnegative rows $U_6, U_{11}, U_{12}, U_{15}$ at this point. A sifted source failing square-freeness has $q^2 \mid a$ for some $q \geq z_*$, and their number is bounded by

$$\sum_{z_* \leq q \leq \sqrt{N}} (N/q^2 + 1) \leq 2N/z_* + \sqrt{N} \leq B.$$

The congruence floor term is retained. Each bad member contributes at most $4 + 2\binom{\omega(a)}{2} \leq 4 + W^2$ to the retained positive rows. Extending negative rows to all of $\mathcal{A}$ only lowers the right side. This proves (8). In particular no three- or four-prime positive multiplicity is charged with a two-prime bound. $\square$

4 Ordinary sieve rows and the exceptional chain

4.1 A common lower envelope

Set $e_1 = 1/35$ and, for $u \geq 1$, define

$$\mathcal{L}(u) = \begin{cases} -4e_1 - (1 + 4e_1)107\eta e^2 e^{-2}, & 1 \leq u < 2, \\ f(u) - 107\eta e^2 h(u) \\ \quad - 3e_1\{F(u) - f(u) + 213\eta e^2 h(u)\}, & u \geq 2. \end{cases} \quad (9)$$

The first branch is negative and is used only as a trivial lower bound. On the second branch f_- is increasing and $\mathcal{F}$ is decreasing; hence $f_- - 3e_1(\mathcal{F} - f_-)$ is increasing and does not exceed f_- . This uses $F - f$ itself, rather than an uncertified numerical maximum.

Lemma 4.1 (Exceptional chain with an exact final mass). *The lower ordinary rows may use (9) at $u = (\theta - \log a/L)/\kappa_1 - \Delta$ in both exceptional alternatives. All upper rows may use a factor $36/35$ and their indicated, possibly smaller, sieve arguments. The normalized finite cost of the exceptional main-mass replacement is at most*

$$100B_E(N) + 1000/L^2 < 10^{-6}, \quad B_E(N) = \frac{96 \log^2(10c)}{\log^\delta x_1(N)}. \quad (10)$$

AP discrepancies are charged separately in (18).

Proof. In the small alternative use [theorem 2.1](#) and the all-modulus estimate (7). For a nonnegative lower model expression, $|\mathcal{A}| > N/L$ may be used; for a negative expression the same lower bound follows instead from the nonnegativity of the actual sifted count. Since $\mathcal{L} \leq f_-$, this gives the stated lower envelope.

In the large alternative write the relevant full conductor as $k = q_1 \cdots q_\ell$ with $q_1 > \cdots > q_\ell$ odd primes, and put $m_j = q_1 \cdots q_j$, $m_0 = 1$, $P^{(j)}(w) = P_N(w)/\prod_{i \leq j} q_i$. Every $q_i < w$, because $k \leq L^{10} < z_*$. The source tags a here have at most two prime factors, all at least z_* , so $(a, k) = 1$. Finite inclusion-exclusion gives, exactly,

$$S(\mathcal{A}_a, P_N(w)) = \sum_{j=0}^{\ell-1} (-1)^j S(\mathcal{A}_{am_j}, P^{(j+1)}(w)) + (-1)^\ell S(\mathcal{A}_{ak}, P^{(\ell)}(w)). \quad (11)$$

This follows by assigning any member not sifted by $P_N(w)$ to the first prime in the ordered list $q_1, \dots, q_\ell$ which fails its coprimality test; the alternating prefix terms cancel.

Use the common level $D_a = N^\theta/(ak^2)$ in every term. The first ℓ model masses are $|\mathcal{A}|/\varphi(am_j)$; their discrepancies are $r(am_j d)$. The last model mass is $|\mathcal{A}_k|/\varphi(a)$; its discrepancies are $r_k(ad)$.

No approximation of the full-conductor prime count is used in applying the relative distribution theorem. In the first terms $k \nmid am_j d$ because $q_{j+1} \nmid d$. In the last term $ad \leq H_N/k$ by (4). All combined moduli are square-free and coprime to N . These are precisely the two AP ranges in (7). Moreover

$$\frac{\log D_a}{\log w} \geq \frac{\theta L - \log a}{\log w} - \Delta.$$

If the right side is below 2, its negative lower envelope follows from $S \geq 0$; no lower linear sieve is applied below its allowed level.

Let $V_j = \prod_{p|P^{(j)}(w)} (1 - 1/(p-1))$ and $\varphi^*(m_j) = \prod_{i \leq j} (q_i - 2)$. The ideal model coefficients satisfy

$$\frac{V_{j+1}}{\varphi(m_j)V_0} = \frac{1}{\varphi^*(m_j)} \left(1 + \frac{1}{q_{j+1} - 2} \right), \quad \frac{V_\ell}{\varphi(k)V_0} = \frac{1}{\varphi^*(k)}.$$

Thus their alternating sum in (11) is exactly one. Their sum for $j \geq 1$ is at most

$$\frac{1}{q_1 - 2} \left(1 + \frac{2}{q_2 - 2} + \frac{2}{(q_2 - 2)(q_3 - 2)} + \cdots \right) \leq \frac{3}{q_1 - 2}.$$

For completeness the tail without the initial one is at most 2: each of its $\ell - 1$ denominator products is at least $q_2 - 2$, and $q_2 \geq 2\ell - 1$, giving $(\ell - 1)/(2\ell - 3) \leq 1$. The case $\ell = 1$ has no tail. The fixed primorial comparison

$$\prod_{2 < p \leq 31} p = 100280245065 < \log^\delta x_1(N_0)$$

implies $q_1 \geq 37$ for every subsequent N : an odd square-free integer all of whose prime factors are at most 31 is at most the displayed primorial. The $j = 0$ local factor is therefore at most $1 + e_1 = 36/35$. Let P_+ and P_- be the sums of the ideal coefficients of the positive and negative terms in (11). The exact identity above gives $P_+ - P_- = 1$, while the tail estimate gives $P_- \leq 3e_1$. Hence

$$f_- P_+ - \mathcal{F} P_- = f_- - P_-(\mathcal{F} - f_-) \geq f_- - 3e_1(\mathcal{F} - f_-) = \mathcal{L}.$$

This accounts explicitly for both the 106η upper-sieve correction and the 107η lower-sieve correction. Applying lower weights to the positive terms and upper weights to the negative terms now gives (9), apart from the final-mass defect and the separately charged AP errors.

The Brun–Titchmarsh inequality and (5) give

$$0 \leq \frac{|\mathcal{A}_k|}{|\mathcal{A}|} \leq \frac{2L}{\varphi(k)(L - \log k)} < \frac{3}{\varphi(k)}.$$

Hence the defect, after multiplication by V_ℓ/V_0 , is at most $3/\varphi^*(k)$. The Rosser–Schoenfeld bound for $k/\varphi(k)$ used by BJS, together with $\log^\delta x_1(N) \leq k \leq L^{10}$, gives $k/\varphi(k) \leq 4 \log(10c)$ here: if $t = \log \log k > 2$, then $e^\gamma t + 2.5/t < (e^\gamma + 5/8)t < 4t$ and $t \leq \log(10c)$. Also

$$\frac{1}{\varphi^*(k)} = \frac{1}{k} \left(\frac{k}{\varphi(k)} \right)^2 \prod_{p|k} \frac{(p-1)^2}{p(p-2)} < \frac{32 \log^2(10c)}{k}.$$

This proves the bound B_E for the defect and for the additional tail-density cost. The reciprocal-prime mass of $[N^{\kappa_1}, N^{\kappa_2})$ is less than $1/3$ by (28); the total available mass is below 3. Each of the two positive pair domains therefore has reciprocal-prime mass below one; $2e^{-\gamma}/\kappa_1 < 14$; and $\mathcal{F}(u) < 2$ for $u \geq 2$. Including the base row of weight four and both pair rows, the defect multiplier after division by four is less than $(4 + 1 + 1) \cdot 14 \cdot 2/4 = 42 < 100$. The local-product endpoints fit the additional $1000/L^2$ in (10). For an upper row, omit q_1 from its prime set. This only enlarges the sifted set, multiplies its model product by at most $36/35$, and removes all multiples of k from its remainder support. This proves the upper assertion as well. $\square$

4.2 Row formulae and the AP multiplicity

Define the universal lower quantities

$$L_0 = \frac{2e^{-\gamma}}{\kappa_1} \mathcal{L}(\theta/\kappa_1 - \Delta), \quad (12)$$

$$L_4 = \frac{2e^{-\gamma}}{\kappa_1} \int_{\kappa_1}^{\kappa_2} \frac{dt}{t} \int_t^{\kappa_2} \mathcal{L}((\theta - t - u)/\kappa_1 - \Delta) \frac{du}{u}, \quad (13)$$

$$L_5 = \frac{2e^{-\gamma}}{\kappa_1} \int_{\kappa_1}^{\kappa_2} \frac{dt}{t} \int_{\kappa_2}^{\lambda-t} \mathcal{L}((\theta - t - u)/\kappa_1 - \Delta) \frac{du}{u}. \quad (14)$$

The upper quantities, before the common exceptional factor, are

$$A_1 = 2e^{-\gamma} \int_{\kappa_1}^{\kappa_2} \mathcal{F}((\theta_- - t)/t) \frac{dt}{t^2}, \quad (15)$$

$$A_2 = \frac{2e^{-\gamma}}{\kappa_1} \int_{\kappa_1}^{\sigma_1} \mathcal{F}((\theta_- - t)/\kappa_1) \frac{dt}{t}, \quad (16)$$

$$A_3 = \frac{2e^{-\gamma}}{\kappa_1} \int_{\kappa_1}^{\sigma_2} \mathcal{F}((\theta_- - t)/\kappa_1) \frac{dt}{t}. \quad (17)$$

For $S(\mathcal{A}_a, P_N(w))$ the ordinary level is N^θ/a . All primes of a are at least w , so $(a, d) = 1$ on the remainder support. The model mass is $|\mathcal{A}|/\varphi(a)$ and its discrepancy is exactly $r(ad)$ in the small or nonexceptional case. Thus these formulae follow from [theorems 2.1](#) and [4.1](#) after the prime-sum transfer proved in [section 7](#). The replacement of $1/\varphi(a)$ by $1/a$, and removal of primes dividing N from lower integrals, are included in the local and finite charges there.

There is no factor equal to the number of source primes in the AP error. For a fixed combined square-free modulus, a tag a of at most two primes can be selected in at most $1 + \omega(d) + \binom{\omega(d)}{2} \leq (1 + L/\log 2)^2$ ways. In the exceptional prefix terms the prefix length $j < \ell$ is uniquely determined by the first absent q_{j+1} , so it adds no factor ℓ . The final prefix uses one fixed relative AP sum. The weights are 4 for the base lower row, 1 for each of the two positive pair rows, and 1 for each of the three upper rows. At most two AP sums occur for a lower row, hence the total multiplier is $2(4 + 1 + 1) + 1 + 1 + 1 = 15 < 16$. Consequently

$$\frac{E_{\text{AP}}}{4\Theta(N)} \leq \frac{16b(N)(1 + L/\log 2)^2}{4(1/2)L} < 10^{-6}. \quad (18)$$

The structural certificate encloses the left side at $c = 24.9$ below $5.623 \cdot 10^{-10}$; the all- c argument is in [section 10](#).

5 The fixed-cutoff rectangle theorem

This section supplies the extension not covered by BJS Lemma 32's printed hypothesis $X > N^{2/3}$. The cutoff below is fixed for all boxes at a given N , not changed to $\log^{10} x_2(Y)$ in the last line.

For $X_3 = N^{1/8}/2$ set $w_3 = \log x_2(X_3)$, $D_0 = w_3^{10}$, and

$$\nu = \frac{100}{L^{\delta/2}(\delta c)^2}, \quad \beta = 1 - \nu.$$

Define, for either $X = X_3$ or $X = z_*/2$, $w_X = \log x_2(X)$ and

$$\begin{aligned} v'(X) &= \frac{3.2 \cdot 10^{-8}}{w_X^4} + w_X^4 \left(\frac{e^{-\nu w_X}}{1 - \nu} + 1.02 w_X^{10} e^{-w_X/2} + 3 w_X^{10} e^{-2w_X/3} \right), \\ v(X) &= v'(X) \left(1 + \frac{1}{\log^{10} X w_X^5} + \frac{1}{(1 - 6/w_X) \log X} \right) + \frac{3}{\log X}. \end{aligned} \quad (19)$$

The BJS maximum over $t \geq x_2(X)$ is bounded by this endpoint value: with $x = \log t$, the functions x^{-4} , $x^4 e^{-\nu x}$, $x^{14} e^{-x/2}$ and $x^{14} e^{-2x/3}$ decrease for $x \geq w_X$, since $\nu w_X > 100$, $w_X > 28$. The bound ν is weaker than both zero bounds used by BJS: all small exceptional thresholds are at most L^δ , and $\nu < 1/(2 \cdot 2.0452 \cdot 10c)$. It therefore works simultaneously for every allowed conductor, after the indicated exceptional deletion.

Theorem 5.1 (Required rectangle extension). *Suppose $N \geq N_0$ is even,*

$$N^{0.398} \leq X \leq N^{7/8}, \quad N^{1/8} \leq Z < Y \leq N, \quad N \leq XY \leq (1 + \varepsilon_0)N,$$

and $|a(n)| \leq 1$. Put $D_R = \sqrt{XY}/\log^{10} Y$. Let k_R be the relevant exceptional conductor up to D_0 . Use all square-free $d < D_R$ coprime to N in the small alternative $k_R < \log^\delta x_2(X_3)$, and only $k_R \nmid d$ in the other alternative. Over either set of moduli,

$$\sum_d \left| \sum_{\substack{n < X \\ np \equiv N \pmod{d}}} \sum_{\substack{Z \leq p < Y \\ (np, d) = 1}} a(n) - \frac{1}{\varphi(d)} \sum_{n < X} \sum_{\substack{Z \leq p < Y \\ (np, d) = 1}} a(n) \right| \leq E_1 + E_2 + E_3 + E_4 + E_5, \quad (20)$$

where the five terms are added, and

$$\begin{aligned} E_1 &= 39v(X_3)XY/\log^3 Y, & E_2 &= 108X \log^{13} Y/\log \log Y, \\ E_3 &= 26XY \log^2 Y/w_3^{10}, & E_4 &= 88XY \log^2 Y(X^{-1/2} + Y^{-1/2}), \\ E_5 &= 106XY/\log^9 Y. \end{aligned}$$

Proof. The hypotheses imply $D_R > 10^9$, $D_0 < D_R$, $D_R < Y^4$, and $Z > X_3$; they also imply $\log \log x_2(X_3) > 10.4$. For the third inequality use $\log D_R \leq L/2 + \log(1 + \varepsilon_0)/2 - 10 \log \log Y < L/2 \leq 4 \log Y$. The other inequalities follow from $\log D_R \geq L/2 - 10c$ and are certified at $c = 24.9$ with the monotone majorants in [section 10](#). These are the precise consequences needed from BJS's original restrictions; no lower bound $X > N^{2/3}$ is used below.

Character orthogonality is valid because $(d, N) = 1$. A character modulo the square-free d has a unique primitive inducing character modulo $r \mid d$; writing $d = rs$ gives $(r, s) = 1$ and exactly one induced character per such primitive character. After removal of the principal character, the left side is at most

$$\sum_{rs < D_R} \frac{\mu^2(rs)}{\varphi(r)\varphi(s)} \sum_{\substack{\chi \bmod r \text{ primitive} \\ \chi \neq \chi_0}} \left| \sum_{\substack{n < X \\ (n, s) = 1}} a(n)\chi(n) \right| \left| \sum_{\substack{Z \leq p < Y \\ p \nmid s}} \chi(p) \right|. \quad (21)$$

All sums may be restricted by the exceptional deletion. Such restrictions are dropped only when this increases a nonnegative majorant.

For $r < D_0$, both prime endpoints are larger than X_3 and $r < \log^{10} x_2(X_3)$, so [\(6\)](#) applies at both. The exact half-open endpoints are $\lceil Y \rceil - 1$ and $\lceil Z \rceil - 1$; the function $t/\log^5 t$ increases here. Thus

$$\sum_{\chi}^* \left| \sum_{\substack{Z \leq p < Y \\ p \nmid s}} \chi(p) \right| \leq 2v(X_3)Y/\log^5 Y + \varphi(r) 1.3841 \log D_R/\log \log D_R.$$

The last term retains every prime dividing s . Bound the integer sum by X and use $\sum_{n \leq D_R} \mu^2(n)/\varphi(n) \leq 1.1 \log D_R$. For $s \geq 16$, BJS's bound $\omega(s) \leq 1.3841 \log s/\log \log s$ is at

most the displayed bound at D_R , since $\log s / \log \log s$ increases there. For $s < 16$, $\omega(s) \leq 2$ is also below that endpoint bound. The result is at most

$$\frac{2.42v(X_3)XY \log^2 D_R}{\log^5 Y} + \frac{1.21 \cdot 1.3841 D_0 X \log^3 D_R}{\log \log D_R} \leq E_1 + E_2.$$

Here $D_0 \leq \log^{10} Y$, $D_R < Y^4$, and $t^3 / \log t$ is increasing for $t = \log D_R$ in this range.

For $r \geq D_0$, split into $R \leq r < 2R$, $R = 2^j D_0$. Cauchy–Schwarz and the multiplicative large sieve in BJS’s proof give

$$\begin{aligned} & \sum_{R \leq r < 2R} \frac{1}{\varphi(r)} \sum_{\chi}^* |A(\chi)P(\chi)| \\ & \leq \frac{1}{R} \{(X + 12R^2)(Y + 12R^2)XY\}^{1/2} \\ & \leq XY/D_0 + \sqrt{12X}Y + \sqrt{12Y}X + 12R\sqrt{XY}. \end{aligned}$$

The support restrictions $(n, s) = 1$ and $p \nmid s$ reduce their square norms and require no extra term. The number of intervals is at most $\log D_R / \log 2$ and their left endpoints sum to at most $2D_R$. Summing also over s yields

$$1.1 \log D_R \left\{ \frac{\log D_R}{\log 2} \left(\frac{XY}{w_3^{10}} + \sqrt{12}XY(X^{-1/2} + Y^{-1/2}) \right) + 24D_R\sqrt{XY} \right\} \leq E_3 + E_4 + E_5.$$

The constants follow from $1.1 \cdot 16 / \log 2 < 26$, $1.1 \cdot 16\sqrt{12} / \log 2 < 88$, and $1.1 \cdot 24 \cdot 4 < 106$. These inequalities include every conductor interval. They prove the claimed fixed-cutoff extension, with no invocation of BJS’s out-of-range printed statement. $\square$

For subsequent aggregation define the slightly larger envelope

$$\begin{aligned} m_N &= 39v(X_3) + \frac{108 \log^{16} X_3}{X_3 \log \log X_3} + \frac{26L^5}{w_3^{10}} \\ &\quad + 88L^5(N^{-0.199} + N^{-1/16}) + \frac{106}{\log^6 X_3}. \end{aligned} \tag{22}$$

Then (20) is at most $m_N XY / \log^3 Y$. The numerator L^5 in the third term is essential because D_0 is fixed. The exponent 0.199 in the fourth term uses the actual new range $X \geq N^{0.398}$, not the old $N^{2/3}$ range. The function $\log^{16} Y / (Y \log \log Y)$ decreases for $Y \geq X_3$; this justifies the second term. The structural certificate gives $m_{N_0} < 1.437 \cdot 10^{-8}$.

The condition audit for this proof is as follows. Each row corresponds to an actual step above; the original condition $X > N^{2/3}$ has no unlisted use.

Step	Required condition	Verification in the new range
Prime endpoints	$Z - 1 > X_3$; $\log \log x_2(X_3) > 10.4$	$Z \geq N^{1/8} = 2X_3$; certified scale
Low conductor	$r < \log^{10} x_2(X_3)$	This is the fixed split $r < D_0$
Exceptional zero	Conductor cut at D_0	Relevant large conductor deleted; common ν otherwise
Induction correction	All primes dividing s counted	Explicit $\omega(s)$ term retained in E_2
Large sieve	Arbitrary complex coefficients of norm at most X, Y	$ a(n) \leq 1$; neither $D_R^2 \leq X$ nor a Type I/II hypothesis is needed
Constants in E_i	$D_R < Y^4$, $D_0 \leq \log^{10} Y$	$Y \geq N^{1/8}$ and $w_3 < \log Y$
Common envelope	$X^{-1/2} \leq N^{-0.199}$, $Y^{-1/2} \leq N^{-1/16}$	Exactly the new lower bounds on X, Y
All boxes	Count and weight each box	The ceiling sum is performed in (24)

6 Five finite box families

We first explain exactly which triple-prime sources suffice for the negative rows. Work on the square-free source from [theorem 3.1](#). In U_7, U_8 , a composite residual would imply $p_1 p_2^3 < N$ with $p_1, p_2 \geq N^{\sigma_2}$, contradicting $4\sigma_2 > 1$. In U_9 it would imply $N^{\kappa_1+3\sigma_1} < N$, and in U_{10} it would imply $N^{\kappa_2+\sigma_2+2\sigma_1} < N$. Both exponent sums exceed one. Consequently the residual is either one or a single prime. A residual equal to one is charged in [equation \(34\)](#). In U_7, U_8, U_9 the third prime is at least p_2 ; in U_{10} it is at least N^{σ_1} , but need not be larger than p_2 . We do not impose that extra ordering.

For each row in the following table choose the indicated selected prime p and the complementary two-prime integer n .

Family	p	exponent interval $[a, b)$	n	Wu weight
7	p_1	$[\sigma_1, 1/3)$	$p_2 p_3$	2
8	p_1	$[\sigma_2, 1/3)$	$p_2 p_3$	2
9h	p_1	$[1/8, \sigma_1)$	$p_2 p_3$	1
9l	p_2	$[\sigma_1, 11/24)$	$p_1 p_3$	1
10	p_3	$[\sigma_1, 1 - \kappa_2 - \sigma_2)$	$p_1 p_2$	1

The two parts of row 9 correspond to $p_1 \geq N^{1/8}$ and $p_1 < N^{1/8}$ respectively. In the latter case $p_2 < \sqrt{N/p_1} \leq N^{11/24}$.

For a family with endpoints a, b put

$$M_{a,b} = \left\lceil \frac{(b-a)L}{\log(1+\varepsilon_0)} \right\rceil, \quad \omega_j = N^a(1+\varepsilon_0)^j, \quad 0 \leq j < M_{a,b}.$$

$$X_j = N/\omega_j, \quad Z_j = \omega_j, \quad Y_j = \min\{(1+\varepsilon_0)\omega_j, N^b\}.$$

Every selected prime lies in exactly one of these half-open intervals, including when the displayed quotient is an integer. The final interval is clipped, not supplemented by an overlapping endpoint box. The coefficient $a_j(n)$ is the indicator of the following set, with $n < X_j$ in every case:

$$\begin{aligned} 7, 8 : & \quad n = p_2 p_3, \quad \omega_j \leq p_2 \leq p_3, \\ 9h : & \quad n = p_2 p_3, \quad N^{\sigma_1} \leq p_2 \leq p_3, \\ 9l : & \quad n = p_1 p_3, \quad z_* \leq p_1 < N^{1/8}, \quad p_3 \geq \omega_j, \quad p_1 \omega_j^2 < N, \\ 10 : & \quad n = p_1 p_2, \quad N^{\kappa_2} \leq p_1 < N^{\sigma_2} \leq p_2, \quad p_1 p_2^2 < N. \end{aligned}$$

Factorization recovers the unordered pair of prime factors of n . In the first three cases the increasing order recovers the labels; in 9l and 10 the disjoint prime ranges recover them. Thus $a_j(n) \in \{0, 1\}$, including the case of a prime square. These conditions do not depend on the selected prime inside its box. The original positive source is contained in this box source, since $pn < N$ implies $\omega_j n < N$.

Define the actual positive integer multiset

$$\mathcal{B}_j = \{|N - pn| : n < X_j, \quad a_j(n) = 1, \quad Z_j \leq p < Y_j\}.$$

Here pn is odd and N is even, so $N - pn \neq 0$. The use of the absolute value preserves divisibility by every modulus and makes the sieve theorem applicable to positive integers. It also permits the overhang $N < pn < (1 + \varepsilon_0)N$ without an unproved strip-distribution estimate.

For the tag (p, n) use model mass one and

$$g_{p,n}(q) = \begin{cases} 0, & q \mid pn, \\ 1/(q-1), & q \nmid pn, \end{cases} \quad (q \nmid N).$$

Then the discrepancy in [theorem 2.1](#) is precisely the expression in (20), since $g_{p,n}(d) = \mathbf{1}_{(pn,d)=1}/\varphi(d)$ for square-free d . There is no substitution of a common density before this identity. Every prime factor of pn is at least z_* and there are at most three, so, for $w = y$,

$$V_{p,n}(y) \leq V_N(y)(1 + 24/z_*). \quad (23)$$

Indeed each deleted factor is $(q-1)/(q-2) \leq 1 + 2/z_*$ and $(1 + 2/z_*)^3 < 1 + 24/z_*$. The same comparison holds after removal of an exceptional $q_1 < z_*$.

Apply [theorem 2.1](#) at $D = N^\theta$, $w = y$. All boxes satisfy

$$N^{0.398} \leq X_j \leq N^{7/8}, \quad Z_j \geq N^{1/8}, \quad N < X_j Y_j \leq (1 + \varepsilon_0)N.$$

By (4), $QD < D_R$ for every box. In the large rectangle-exceptional case delete its largest prime q_1 . Its threshold exceeds $\prod_{2 < p \leq 31} p$, so $q_1 \geq 37$ and the product inflation is at most $36/35$. No equality between this exceptional conductor and the one in the ordinary rows is assumed. The argument $3\theta - \Delta$ is smaller than the actual argument 3θ , hence is safe in either case.

Writing the table's weights as $\lambda_{a,b}$, the complete rectangle remainder is therefore bounded by

$$\frac{E_{\text{rect}}}{4\Theta(N)} \leq \frac{(1 + \varepsilon_0)m_N}{4(1/2)L} \sum_{\text{families}} \frac{\lambda_{a,b}}{a^3} \left(\frac{(b-a)L}{\log(1 + \varepsilon_0)} + 1 \right) < 0.0085. \quad (24)$$

The evaluated left side is below 0.008032 at $c = 24.9$. This is a sum over all five families and every box, not a per-box error. In particular, there is no implicit factor six in the coefficient norm.

We record the main integrals. Counting the remaining third prime by

$$\pi \left(\frac{(1 + \varepsilon_0)N}{p_1 p_2} \right) \leq \frac{1.00005(1 + \varepsilon_0)N}{L p_1 p_2 (1 - t - u)}, \quad t = \log p_1 / L, \quad u = \log p_2 / L,$$

is legitimate throughout these boxes: $1 - t - u > 1/4$ and the logarithm of every prime-counting argument is at least $L/4 > 30002$. In particular $(1 - 6/L)^{-1} < 1.00005$, as checked by the certificate. The changes in the exponent domains are bounded in [section 7](#). At the unbroadened domain define

$$J_7 = \int_2^{1/\sigma_1-1} \frac{\log(v-1)}{v} dv, \quad J_8 = \int_2^{1/\sigma_2-1} \frac{\log(v-1)}{v} dv, \quad (25)$$

$$J_9 = \int_{\kappa_1}^{\sigma_1} \frac{\log((1-t-\sigma_1)/\sigma_1)}{t(1-t)} dt, \quad J_{10} = \int_{\kappa_2}^{\sigma_2} \frac{\log((1-t-\sigma_2)/\sigma_2)}{t(1-t)} dt. \quad (26)$$

For example, the inner integral in rows 7, 8 is $\int_t^{(1-t)/2} du/[u(1-t-u)] = \log((1-2t)/t)/(1-t)$; the change $v = 1/t - 1$ gives (25). For rows 9, 10 replace the lower inner endpoint by σ_1, σ_2 respectively, giving (26). The upper main quantities used in the ledger are

$$B_i = \frac{36}{35} 6e^{-\gamma} \mathcal{F}(3\theta - \Delta) J_i \quad (i = 7, 8, 9, 10). \quad (27)$$

The factor $1.00005(1 + \varepsilon_0)$, local products, and finite deletions have not been hidden in these integrals; they are charged separately below.

7 Finite transfers and local terms

7.1 A prime-measure estimate with explicit variation

Let μ_N be the measure on $[\kappa_1, 1]$ putting mass $1/p$ at $\log p/L$ for $z_* \leq p < N$. Let $d\mu(t) = dt/t$ there. BJS's reciprocal-prime inequalities, including the at-most-one-prime endpoint correction, give

$$\sup_{\kappa_1 \leq t \leq 1} |(\mu_N - \mu)([\kappa_1, t])| \leq d_N := \frac{3 \cdot 10^{-6}}{\kappa_1 L} + 2N^{-\kappa_1}, \quad \mu_N([\kappa_1, 1]) < 3. \quad (28)$$

The constant $3 \cdot 10^{-6}$ bounds $2.964 \cdot 10^{-6} + 1.436 \cdot 10^{-16}$, not twice the sum: the two endpoint errors each lie in the same asymmetric interval, so their difference has at most that interval's length.

Here is the finite-dimensional transfer we use. Suppose the coordinate sections of a domain are unions of at most ten intervals, and its kernel G has magnitude at most M , at most ten additional jump points in each section, and derivative at most M' away from those points. Integration by parts against the cumulative discrepancy in (28) bounds a one-coordinate difference by $d_N(40M + M')$. The coefficient $40M$ includes both domain endpoints and both sides of every jump. Telescoping the product measures gives

$$\left| \int G d\mu_N^d - \int G d\mu^d \right| \leq d 3^{d-1} d_N(40M + M'). \quad (29)$$

To see the telescoping explicitly, replace coordinates $1, 2, \dots, d$ one at a time. Condition on the other $d - 1$ coordinates, apply the one-dimensional integration-by-parts bound, and integrate that bound against their product measure, whose mass is at most 3^{d-1} . This proof also covers strict inequalities and diagonal atoms.

The following bounds apply to the kernel excluding the factors dt/t . They include the displayed sieve prefactors, but not the Wu row weight.

Kernel	d	M	M'
A_1	1	120	20000
A_2, A_3 (each)	1	120	2000
L_4, L_5 (each)	2	48	3000
Each triple-prime box mass	2	128	512
Each switched four-prime mass	4	8000	0

These are analytic bounds, not values sampled on a mesh. On the domains at issue, $1/t \leq 12$, $1/t^2 \leq 144$, $F \leq 4$, $f \leq 1$, $|F'|, |f'| \leq 5$, and $|h'| \leq 1$. In fact $\mathcal{F} \leq 4$ and $|\mathcal{F}'| \leq 6$: the first follows from $2e^\gamma + 106\eta < 4$, and the second from $106\eta e^2 < 1$. These inequalities are included in the certificate. The delay equations prove the derivative bounds directly, using $s \geq 1$ (and $s - 1 \geq 1$ whenever its nonzero term is needed). For $\mathcal{L}$, $|\mathcal{L}| \leq 2$ and $|\mathcal{L}'| \leq 10$ on its smooth pieces. The arguments in A_1 have derivative at most 72, and the other ordinary arguments at most 12; the prefactor $2e^{-\gamma}/\kappa_1 < 14$ then gives the first three lines. In a box kernel, $1 - t - u > 1/4$ and $(36/35)6e^{-\gamma}\mathcal{F}(3\theta - \Delta) < 16$, giving $M < 64$, $M' < 256$; the table doubles these to cover the inflated box. The switched multiplier is less than 8000 using $2e^{-2\gamma}/\kappa_1^2 < 100$ and $F \leq 4$. All boundaries are linear exponent inequalities; their sections are intervals. Breaks of the sieve functions at 2, 3, 4, 5, 6 and the jump of $\mathcal{L}$ at 2 use fewer than ten additional points. These facts prove every entry of the table.

The sum of the right sides of (29), including the Wu weights and the two switched rows, is less than $10^9 d_N$ even before division by four. We allow $10^{10} d_N$ in the normalized ledger.

7.2 Box overhangs, prime counting and products

Put $h_0 = \log(1 + \varepsilon_0)/L$. For rows 7, 8, the box condition gives $p_2 \geq p_1/(1 + \varepsilon_0)$ and $p_1 p_2^2 < (1 + \varepsilon_0)N$. Thus the exponent boundaries move from $u = t$ to $u \geq t - h_0$ and from $t + 2u < 1$ to $t + 2u < 1 + h_0$. For 9h the lower condition $u \geq \sigma_1$ is unchanged and only the latter boundary moves. For 9l, $p_3 \geq p_2/(1 + \varepsilon_0)$ and $p_1 p_2^2 < (1 + \varepsilon_0)^2 N$; the upper boundary moves by at most $2h_0$. For 10, $p_1 p_2^2 < N$ is retained exactly and the product bound only enlarges to $p_1 p_2 p_3 < (1 + \varepsilon_0)N$. The $p_1 = N^{1/8}$ split is disjoint and makes no additional mass. The five families consequently require at most eight strips, each of coordinate width at most $2h_0$. Their continuous mass is bounded using $dt/t \leq 12 dt$ and the table. Their discrete mass first uses (29), already budgeted above. A common conservative bound for all these continuous strips is $10^{10} \cdot 16h_0$.

The raw total of all negative main terms in the final ledger is below 60, certified with Arb. Since $1.00005(1 + \varepsilon_0) < 1.0002$, the common prime-count inflation costs at most $60/(4 \cdot 5000) = 0.003$. Therefore

$$E_{\text{transfer}}/(4\Theta(N)) \leq 10^{10}(d_N + 16h_0) + 0.003 < 0.0035. \quad (30)$$

This also permits the factor 1.00005 in ordinary upper model masses. The evaluated proxy is less than 0.003251 at $c = 24.9$.

The product calculation in BJS Lemma 38, or its displayed factorization into the full Mertens product and the local factors of N , gives

$$V_N(N^t) = \frac{2e^{-\gamma}C_N}{tL}(1 + \epsilon_t), \quad |\epsilon_t| \leq \frac{2}{(\kappa_1 L)^2} + 10LN^{-\kappa_1}, \quad t \geq \kappa_1. \quad (31)$$

To specify the tail used here: the full prime product contributes at most $1/\log^2(N^t)$; the tail of $\prod_{p>2}(1 - 1/(p-1)^2)$ is bounded by $2/(N^t - 2)$; and the omitted local factors with $p \mid N$, $p \geq N^t$ are bounded in logarithm by $2\omega(N)/(N^t - 2)$. Using $\omega(N) \leq L/\log 2$ and $e^x \leq 1 + 2x$ for $0 \leq x \leq 1/2$ gives (31). The same bound covers all prefix products, with $\log(Nm_j) \leq L + 10c$; their local ratios are already retained exactly in [theorem 4.1](#). The full integer-sieve product satisfies

$$\prod_{p < z_*} (1 - 1/p) \leq (1 + \eta)e^{-\gamma}/\log z_*. \quad (32)$$

For example BJS's quoted Mertens bound $e^{-\gamma}(1 + \log^{-2} z_*)/\log z_*$ is stronger, since $(\kappa_1 L)^{-2} < \eta$. There is no N -dependent local factor in (32).

For a tag of one or two primes at least z_* , $a/\varphi(a) \leq (1 - 1/z_*)^{-2} < 1 + 4/z_*$. Together with (23), (31), and the total kernel bounds above, all such normalized local corrections are bounded by

$$E_{\text{local}}/(4\Theta(N)) \leq 1000 \left(\frac{2}{(\kappa_1 L)^2} + 40LN^{-\kappa_1} + L^{-9} \right) < 10^{-6}. \quad (33)$$

The L^{-9} allowance here is harmless slack, not a second charge of the integer rough-sieve remainder, which is assigned to the switch. In lower rows where the chosen lower function is negative, use $S \geq 0$ before making any mass or product replacement.

7.3 A single finite combinatorial envelope

All combinatorial exclusions used here are bounded, before division by $4\Theta(N)$, by

$$\begin{aligned} E_{\text{finite}} := & (4 + 7W^2)B + 6W^2N^{2/3} + 24N^{1/3} \\ & + 4(1 + W)^5N^{\kappa_1} + 100(1 + W)^6N^{1-\kappa_1}, \quad W = L/\log 2. \end{aligned} \quad (34)$$

Here are the assignments, so no unspecified terminal is needed. The first $(4 + W^2)B$ is [theorem 3.1](#). When rows 7–10 are converted to three-prime sources, the removed nonsquare-free source members contribute at most $6W^2B$: each tag uses two distinct divisors and the four Wu weights sum to six. A residual equal to one gives $p_1p_2 < N^{2/3}$ in all four rows, so at most $6W^2N^{2/3}$ is lost. For a complementary prime below y , there are at most y source values and at most four ordered triple tags per value; after the weights this is at most $24y$.

In rows 13, 14, a complementary prime below z_* has at most $\binom{\omega(a)}{4} \leq (1 + W)^4$ ordered tags for its source $a = N - p$. Allowing also the at most W complementary primes dividing N gives less than $4(1 + W)^5z_*$. The latter allowance is unnecessary for our definition of $\mathcal{A}$, but explicitly covers the more general local implication in [theorem 8.1](#). Finally, removing a prime $p \mid N$, $p \geq z_*$ from a lower prime integral costs a reciprocal mass at most W/z_* . Multiplying by the kernel bounds, the other coordinate masses, N/L^2 , and all Wu weights is less than the last term in (34). That term also covers any endpoint prime or repeated-prime boundary that is removed before

using an ordered source. Ordered-source diagonals in the integral transfers are already allowed by (29), so this is conservative. All floor contributions in these counts use $N/q^2 + 1$, as in B ; they have not been replaced by N/q^2 . Since $C_N > 1/2$, direct comparison of the five displayed terms gives

$$E_{\text{finite}}/(4\Theta(N)) \leq e^{30} L^{12} N^{-1/12} < 10^{-6}. \quad (35)$$

The explicit logarithmic comparison and its half-line validity are included in the certificate and section 10.

8 The complementary-prime switch

8.1 Local implication and unique factor recovery

Let $\mathcal{C}_{13}, \mathcal{C}_{14}$ be the multisets of tuples $(p_1, p_2, p_3, p_4, r_0)$ satisfying the respective prime ranges in U_{13}, U_{14} , and

$$r_0 \geq 1, \quad (r_0, P_{\text{all}}(p_2)) = 1, \quad p_1 p_2 p_3 p_4 r_0 \leq N - 2.$$

Do not impose coprimality with N in these enlarged sets. Set $b = N - p_1 p_2 p_3 p_4 r_0$, so b is a positive odd integer. Tags are retained even if their values b coincide.

Lemma 8.1 (The local switch, with its exceptional alternative). *Write $\varpi = N - p_1 p_2 p_3 p_4 r_0$ and suppose the four primes are coprime to N . If ϖ is prime and $(r_0, P_N(p_2)) = 1$, then either $\varpi \mid N$ or $(r_0, N) = 1$ and $(r_0, P_{\text{all}}(p_2)) = 1$. For $\varpi \geq z_*$ in the latter case, the complementary upper Rosser weight at z_* is at least one.*

Proof. If a prime q divides (r_0, N) , it divides ϖ , hence $q = \varpi \mid N$. Otherwise $(r_0, N) = 1$; every prime below p_2 dividing N is then absent from r_0 , and every other such prime is absent by the original sift. This is precisely full p_2 -roughness. The upper Rosser divisor sum bounds the sifted indicator pointwise; a prime $\varpi \geq z_*$ has no prime factor below z_* . Its divisor sum is therefore at least one, and is nonnegative on every positive integer. This last fact permits enlargement to $\mathcal{C}_i$ after the implication has been applied. $\square$

For $\mathcal{A}$ itself $\varpi \nmid N$, so only the main alternative occurs. Apart from the finite small-prime charge in (34), U_i is bounded by the sum of the upper Rosser weights on $\mathcal{C}_i$. We retain full p_2 -roughness throughout the distribution proof. It is weakened to z_* -roughness only when bounding the principal model mass.

Regroup the source by

$$n = p_3 p_4, \quad m = p_1 p_2 r_0.$$

The two prime factors of n determine (p_3, p_4) , since $p_3 < p_4$. The two least prime factors of m , counted with multiplicity, determine (p_1, p_2) : $p_1 < p_2$ and every prime factor of r_0 is at least p_2 . Dividing m by $p_1 p_2$ then recovers r_0 . Thus both coefficient sequences are exactly bounded by one. This remains true when $p_2 \mid r_0$ or $r_0 = 1$; no square-free condition on r_0 is needed. The only cross-conditions between the two groups are

$$p_2 < p_3, \quad mn \leq N - 2. \quad (36)$$

All other prime ranges, including $p_3 p_4 < N^\lambda$ in row 14, belong to one group. In particular $N^{1/6} \leq n < N^\lambda < N^{1/3}$.

Here are the coefficient formulas that will be used below. Let

$$\begin{aligned} \mathcal{N}_{13} &= \{(p_3, p_4) : z_* \leq p_3 < p_4 < N^{\kappa_2}\}, \\ \mathcal{N}_{14} &= \{(p_3, p_4) : z_* \leq p_3 < N^{\kappa_2} \leq p_4, p_3 p_4 < N^\lambda\}, \\ \mathcal{M} &= \{(p_1, p_2, r_0) : z_* \leq p_1 < p_2 < N^{\kappa_2}, r_0 \geq 1, (r_0, P_{\text{all}}(p_2)) = 1\}. \end{aligned}$$

For a dyadic block $I_X = [X, 2X)$, clipped at $N^{1/3}$, put $M_X = N/X$ and define

$$\begin{aligned}\alpha_{i,X}(n) &= \#\{(p_3, p_4) \in \mathcal{N}_i : p_3 p_4 = n, n \in I_X\}, \\ \beta_X(m) &= \#\{(p_1, p_2, r_0) \in \mathcal{M} : p_1 p_2 r_0 = m, m \leq M_X\}.\end{aligned}$$

The unique recovery just proved says, pointwise,

$$\alpha_{i,X}(n), \beta_X(m) \in \{0, 1\}, \quad \sum_n |\alpha_{i,X}(n)|^2 < 2X, \quad \sum_m |\beta_X(m)|^2 \leq M_X.$$

On their supports denote the recovered primes by $p_3(n)$ and $p_2(m)$. Because $n \geq X$, a pair with $m > M_X$ cannot pass $mn \leq N - 2$; therefore the cutoff in β_X loses nothing. For every character χ , the following is an exact finite identity:

$$\sum_{\mathcal{C}_i} \chi(mn) = \sum_X \sum_{n,m} \alpha_{i,X}(n) \beta_X(m) \chi(nm) \mathbf{1}_{p_2(m) < p_3(n)} \mathbf{1}_{mn \leq N-2}. \quad (37)$$

The blocks are disjoint, so this identity contains neither an ordering multiplicity nor a hidden convolution energy estimate.

8.2 Exact character decomposition and the low conductors

Use $D_\circ = N^{2/5}$, $w = z_*$ and $D_* = QD_\circ$. For the switched exceptional alternative use the conductor cutoff

$$X_q = z_*/2, \quad w_q = \log x_2(X_q), \quad q_0 = \lceil w_q^{10} \rceil - 1.$$

Thus $w_q^{10}/2 \leq q_0 < w_q^{10}$, including at integral values of w_q^{10} . If the relevant conductor up to q_0 is at least w_q^δ , remove its largest prime q_1 from the sieve prime set. The threshold exceeds $\prod_{2 < p \leq 31} p$ and hence its largest prime is at least 37; its product cost is at most 36/35. In the other case make no deletion. The zero bound ν from (19) is valid in both cases.

Theorem 8.2 (Finite switched distribution). *For the two explicitly defined tuple multisets $\mathcal{C}_i$, use all square-free $d < D_*$ with $d \mid P_N(z_*)$ in the small switched alternative, and omit multiples of q_1 in the large alternative. For the discrepancies $r_i(d)$ defined in (38),*

$$\sum_d (|r_{13}(d)| + |r_{14}(d)|) \leq E_{\text{low}} + E_{\text{high}} + E_P,$$

with the complete formulas (40), (43) and (42). Together with (44) this proves the switched upper main term (45), with only the transfers, local products and finite exclusions explicitly assigned in section 7.

The proof follows; in particular, the theorem does not assume a weighted bilinear distribution result.

Every source factor of mn is at least z_* ; every prime factor of a supported modulus is below z_* . Consequently $(mn, d) = 1$ identically on the support. Character orthogonality gives the exact remainder

$$\begin{aligned}r_i(d) &= \#\{(m, n) \in \mathcal{C}_i : mn \equiv N \pmod{d}\} - |\mathcal{C}_i|/\varphi(d) \\ &= \frac{1}{\varphi(d)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{(m,n) \in \mathcal{C}_i} \chi(mn).\end{aligned} \quad (38)$$

For square-free $d = rs$, $(r, s) = 1$, each character has exactly one primitive inducing character modulo r . Since $(mn, s) = 1$, induction introduces no correction on this source. In particular the often missing term $p_4 \mid s$ is identically zero, not merely small. Therefore

$$\sum_{d < D_*} |r_i(d)| \leq \sum_{rs < D_*} \frac{\mu^2(rs)}{\varphi(r)\varphi(s)} \sum_{\substack{\chi \bmod r \text{ primitive} \\ \chi \neq \chi_0}} \left| \sum_{\mathcal{C}_i} \chi(p_1 p_2 p_3 p_4 r_0) \right|. \quad (39)$$

The modulus set retains $(d, N) = 1$ and the exceptional deletion. We drop such restrictions only inside nonnegative upper bounds.

For $r \leq q_0$, fix p_1, p_2, p_3, r_0 first. The allowed p_4 form a single interval with lower endpoint at least z_* and upper endpoint at most $N/(p_1 p_2 p_3 r_0)$. For row 13 intersect (p_3, N^{κ_2}) with that upper bound; for row 14 intersect $[N^{\kappa_2}, N^\lambda/p_3]$ with it. Empty intervals contribute zero. Both prime-counting endpoints, rounded to the corresponding integers, exceed X_q . BJS Lemma 29 applies before any Perron integral is introduced. Its bound at the two endpoints, and $\log p_4 \geq \kappa_1 L$, give at most

$$\frac{2v(X_q)N}{(\kappa_1 L)^5 p_1 p_2 p_3 r_0}$$

for the sum of absolute primitive-character prime sums. The reciprocal-prime mass of $[N^{\kappa_1}, N^{\kappa_2})$ is less than $1/3$ by (28). The ordered three-prime reciprocal sum is at most its cube divided by 6, and is therefore below one. Also $\sum_{r_0 \leq N} 1/r_0 \leq 1 + L$. Finally $\sum_{r \leq q_0} \mu^2(r)/\varphi(r) \leq 1.1 \log(w_q^{10})$ and $\sum_{s < D_*} \mu^2(s)/\varphi(s) \leq 1.1 \log D_*$. Thus the two rows together satisfy

$$E_{\text{low}} \leq \frac{4v(X_q)N(1+L)}{(\kappa_1 L)^5} 1.21 \log(w_q^{10}) \log D_*. \quad (40)$$

This is a bound for all low conductors and both variable endpoints. At $c = 24.9$ it is below $1.076 \cdot 10^{-12} \Theta(N)$ using the uniform envelope $v(X_q) \leq 40/L$.

8.3 Perron separation, with an explicit tail

Divide n into intervals $X \leq n < 2X$, beginning at $N^{1/6}$ and clipping the last at $N^{1/3}$. There are at most $K_n = 2 + L/(6 \log 2)$ intervals. On one interval write $M = M_X = N/X$ and abbreviate the coefficients in (37) as $\alpha(n), \beta(m)$. The two exact tests in (36) are

$$\mathbf{1}_{p_3/(p_2+1/2) > 1}, \quad \mathbf{1}_{(N-3/2)/(mn) > 1}.$$

They avoid equality at the integration boundary for every integer tuple. Put $\sigma = 1/L$, $T = N^4$. For $x > 0$, $x \neq 1$, let

$$\mathcal{P}_T(x) = \frac{1}{2\pi i} \int_{\sigma-iT}^{\sigma+iT} \frac{x^s}{s} ds.$$

Contour integration to the right or left of the pole at zero, followed by integration by parts on the two horizontal rays, gives

$$|\mathcal{P}_T(x) - \mathbf{1}_{x>1}| \leq \frac{4x^\sigma}{T|\log x|}. \quad (41)$$

For precision, the untruncated vertical integral equals the indicated residue; for a tail ray $\int_T^\infty e^{it \log x}/(\sigma + it) dt$, integration by parts bounds its absolute value by $2/(T|\log x|)$. The two rays and $1/(2\pi)$ give the weaker constant 4 above.

Both ratios lie between $1/(2N)$ and $2N$, have $|\log x| \geq 1/(4N)$, and $x^\sigma \leq e^2$. The latter two facts follow from their half-integer gaps and all participating integers being at most $2N$. Their individual tails are at most $16e^2 N/T$. As $16e^2 N/T < 1$, replacing the product of the two indicators by the product of the two integrals costs at most $64e^2 N/T$ per pair. There are at most $2XM = 2N$ pairs per block, fewer than D_* moduli, and two rows; after summing the character weights $\varphi(d)^{-1} \sum_x 1 \leq 1$, the full tail is at most

$$E_P \leq 256e^2 K_n N^2 D_*/T. \quad (42)$$

It is bounded after normalization by the even larger envelope $e^{20} L^{20} D_* N^{-2}$ in the certificate.

The two remaining integrals separate as a product of a sum in n and a sum in m . Their weights are $p_3^s n^{-t}$ and $(p_2 + 1/2)^{-s} m^{-t}$, with the scalar $(N - 3/2)^t$ outside. Here $\Re s = \Re t = \sigma$.

Each weight is bounded by e^2 , and the outside scalar by e . Consequently the square norms and the outside factor together cost less than e^6 , uniformly in both imaginary parts. The total variation of each Perron kernel is

$$\frac{1}{2\pi} \int_{-T}^T \frac{dt}{|\sigma + it|} = \frac{1}{\pi} \operatorname{arsinh}(T/\sigma) < 2L.$$

The certificate verifies the endpoint inequality; its difference is increasing on the half-line. No maximal-sum theorem or complex small-conductor prime estimate is being assumed.

8.4 The four large-conductor terms are all added

For $r > q_0$ split into $R \leq r < 2R$ and use the multiplicative large sieve. Zero-pad the n -sequence to the prefix $n < 2X$. Then

$$(2X + 12R^2) \sum_n |\alpha(n)|^2 < (2X + 12R^2)2X \leq 4(X + 12R^2)X,$$

whereas the m -sequence contributes at most $(M + 12R^2)M$. Thus the square root costs the explicit factor 2 in the next display; it also absorbs every endpoint $+1$. Cauchy–Schwarz, $|\alpha|, |\beta| \leq 1$, and the weights just bounded therefore give the following majorant before the two Perron factors:

$$2e^6 \frac{\{(X + 12R^2)(M + 12R^2)XM\}^{1/2}}{R}.$$

In particular, the floor $+1$ is already covered by the norm $\sum_{n < 2X} |\alpha(n)|^2 < 2X$ and $\sum_{m \leq M} |\beta(m)|^2 \leq M$. The elementary inequality

$$\frac{\{(X + 12R^2)(M + 12R^2)XM\}^{1/2}}{R} \leq \frac{XM}{R} + \sqrt{12X} M + \sqrt{12M} X + 12R\sqrt{XM}$$

has four positive terms. Summing $1/R$ geometrically gives at most $2/q_0$; summing R gives at most $2D_*$; the number of intervals is at most $K_R = 2 + 0.4154L/\log 2$. Define

$$B_1 = 2XM/q_0, \quad B_2 = K_R\sqrt{12X} M, \quad B_3 = K_R\sqrt{12M} X, \quad B_4 = 24D_*\sqrt{XM}.$$

After both rows, all n blocks, both Perron factors, and the s -sum in (39), a conservative bound is

$$E_{\text{high}} \leq 64e^6 K_n (2L)^2 (1.1 \log D_*) (B_1 + B_2 + B_3 + B_4). \quad (43)$$

In this expression $XM = N$, $X \geq N^{1/6}$, $M \geq N^{2/3}$, $q_0 \geq w_q^{10}/2$, and $D_* < N^{0.4154}$. These substitutions give a quantity below $10^{-20}\Theta(N)$ for every $c \geq 24.9$; at the endpoint the certificate encloses it below $1.597 \cdot 10^{-28}\Theta(N)$. Induced characters have been counted once through $d = rs$, and there is no uncharged ordering multiplicity.

8.5 The integer rough sieve and the switched main mass

Only now replace the condition on r_0 by full z_* -roughness. For $P = p_1 p_2 p_3 p_4$ let $R = N/P$, enlarging the upper endpoint $(N - 2)/P$ when necessary. In both rows

$$R \geq N^a, \quad a = 1 - \lambda - 2\kappa_2 = \frac{1}{2} + 2\kappa_1 - 2\kappa_2 = \frac{999}{2000}.$$

For row 13 this follows from $p_1 p_2 p_3 p_4 < N^{4\kappa_2}$, which is stronger. For row 14 it follows directly from $p_3 p_4 < N^\lambda$ and $p_1 p_2 < N^{2\kappa_2}$. Use the integer sequence $1 \leq r_0 \leq R$, model total R , density $g(q) = 1/q$, and

$$D_r = \frac{R}{Q_{\text{all}} \log^{10} R}, \quad Q_{\text{all}} = \prod_{p \leq 10^9} p.$$

For every d , the discrepancy is $\lfloor R/d \rfloor - R/d$ and its absolute value is at most one. There are fewer than $Q_{\text{all}} D_r = R/\log^{10} R$ supported positive integers d . Moreover

$$\frac{\log D_r}{\log z_*} \geq \frac{aL - \bar{q} - 10 \log(aL)}{\kappa_1 L} > s_r := \frac{12037}{2500}.$$

The monotonicity of $x - 10 \log x$ for $x > 10$ proves the first inequality. The second holds uniformly by [section 10](#). The model sieve and (32) give

$$\#\{r_0 \leq R : (r_0, P_{\text{all}}(z_*)) = 1\} \leq \frac{(1 + \eta)e^{-\gamma}}{\kappa_1 L} \mathcal{F}(s_r) R + \frac{R}{\log^{10} R}. \quad (44)$$

This is a statement about the full prime product, not $P_N(z_*)$.

The complementary model is $|\mathcal{C}_i|$ with common density $1/(q - 1)$. The upper model sieve at D_o has argument $\log D_o / \log z_* = 800/191$. To use a single conservative expression in both exceptional alternatives, replace it by $800/191 - \Delta$ and include $36/35$. Summing the main term of (44) over the prime ranges and applying the already budgeted transfer gives

$$B_{\text{sw}} = \frac{36}{35} (1 + \eta) \frac{2e^{-2\gamma}}{\kappa_1^2} \mathcal{F}(s_r) \mathcal{F}(800/191 - \Delta) (I_{13} + I_{14}), \quad (45)$$

where

$$I_{13} = \frac{\log^4(\kappa_2/\kappa_1)}{24}, \quad I_{14} = \int_{\kappa_1}^{\kappa_2} \frac{\log^2(t/\kappa_1)}{2t} \log\left(\frac{\lambda - t}{\kappa_2}\right) dt.$$

Indeed the first is the ordered four-simplex of the measure dt/t . In the second, fixing $t_3 = t$ gives $\int_{\kappa_1 < t_1 < t_2 < t} dt_1 dt_2 / (t_1 t_2) = \frac{1}{2} \log^2(t/\kappa_1)$, and then integrate $\kappa_2 \leq t_4 < \lambda - t$.

The remainder of (44), after the complementary sieve multiplier, is at most $e^{20} L^{-9} \Theta(N)$. For a direct bound, its reciprocal four-prime sum is below $3^4 = 81$, $\log R \geq aL$, $V_N(z_*) \leq 14C_N/L$, and $(36/35) \mathcal{F}(800/191 - \Delta) < 3$. These yield $3402a^{-10} L^{-9} \Theta(N) < e^{20} L^{-9} \Theta(N)$. All constants here are checked as rational/Arb inequalities.

Combining (40), (42), (43), and this last rough remainder proves the required complete distribution and switching estimate. Their sum, even without using the final division by four, is below $10^{-6} \Theta(N)$. The ledger uses the allowance

$$E_{\text{sw}} / (4\Theta(N)) < 10^{-6}. \quad (46)$$

Small complementary primes and local source exclusions remain assigned to (34); the switch error is not counted twice.

9 One forward ledger and the assembly

All the numbers in this section are enclosed by the Arb computation `c249_verify.py`; no decimal is fitted to a desired final answer. The symbols in the first column are exactly those in (12)–(27) and (45). The following intervals are rounded outwards to twelve decimal places. The signed weight is applied before the final division by four.

Quantity	Lower endpoint	Upper endpoint	Signed Wu weight
L_0	11.738503274673	11.738503274674	4
A_1	0.031268140768	0.031268140769	−36/35
A_2	16.368118871827	16.368118871828	−36/35
A_3	16.357231429105	16.357231429106	−36/35
L_4	0.000032859562	0.000032859563	1
L_5	0.011400175493	0.011400175494	1
B_7	0.164463777759	0.164463777760	−2
B_8	0.166078002505	0.166078002506	−2
B_9	6.263354678712	6.263354678713	−1
B_{10}	6.254338521550	6.254338521551	−1
B_{sw}	0.000000172621	0.000000172622	−1

The box quantities and B_{sw} already include $36/35$; applying that factor to them a second time would be incorrect. The three A_i do not include it. The single master expression is

$$\mathfrak{M} = \frac{4L_0 - (36/35)(A_1 + A_2 + A_3) + L_4 + L_5 - 2B_7 - 2B_8 - B_9 - B_{10} - B_{\text{sw}}}{4}. \quad (47)$$

Positive rows use lower interval endpoints and negative rows use upper ones. Even using only the printed twelve-place intervals gives $\mathfrak{M} > 0.0235368436443 > 0.0228$. Using the full balls gives

$$\mathfrak{M} \in [0.0235368436464714387690418419462, 0.0235368436464714387690418419466].$$

The auxiliary small-exceptional computation is not needed for the theorem. Our universal worst-case expression is used even if the ordinary, rectangular and switched exceptional choices are different.

The following are *allocated upper bounds*, already normalized by $4\Theta(N)$. Each actual error was bounded before this allocation.

Error family	Defining bound	Allocated upper bound
All rectangle boxes	(24)	0.008500
Prime measures, box boundaries, prime-count inflation	(30)	0.003500
Ordinary AP remainders	(18)	0.000001
Exceptional full-conductor mass	(10)	0.000001
Switch distribution, Perron, integer rough floor	(46)	0.000001
Finite combinatorial exclusions	(35)	0.000001
Local sieve products	(33)	0.000001
Total		0.012005

There is one charge per error family. In particular the integer rough floor appears in the switched error and the square-divisor source term appears in the finite error; neither is subtracted again.

Applying [theorem 3.1](#), the ordinary model estimates, the five box estimates and the switched estimate, followed by their transfers, gives for every even N in the range where the seven bounds hold

$$\frac{D_{1,2}(N)}{\Theta(N)} \geq \mathfrak{M} - \frac{E_{\text{rect}} + E_{\text{transfer}} + E_{\text{AP}} + E_{\text{exc}} + E_{\text{sw}} + E_{\text{finite}} + E_{\text{local}}}{4\Theta(N)}. \quad (48)$$

The errors here denote their assigned total upper envelopes, so an unused allowance is still subtracted. The next section proves that every envelope is valid on the full half-line. It follows that the right side of (48) is greater than $0.0228 - 0.012005 = 0.010795 > 0$, proving [theorem 1.1](#). This internal positive margin is not an attempt to optimize an explicit coefficient.

10 Uniformity on the entire half-line

In this section c varies continuously over $[24.9, \infty)$ and $L = e^c$. Establishing bounds on this larger continuous range suffices for every even integer N . The certificate checks only the starting inequalities used below; all propagation is analytic.

10.1 Fixed levels and exceptional alternatives

The function

$$\frac{1}{2} - e^{-c}(\bar{q} + 20c)$$

has derivative $e^{-c}(\bar{q} + 20c - 20) > 0$. The exceptional level loss $20ce^{-c}/\kappa_1$ has negative derivative. For $a = 999/2000$, the derivative of

$$\frac{a - e^{-c}\{\bar{q} + 10\log(ae^c)\}}{\kappa_1}$$

is $e^{-c}\{\bar{q} + 10\log(ae^c) - 10\}/\kappa_1 > 0$. These facts prove all uses of θ, Δ, s_r on the half-line. Also $\log D_*/L \leq 2/5 + \bar{q}/L < 0.4154$ everywhere. The functions $\log x_1(N) = L - 5c$ and $\log x_2(N^b/2) = bL - \log 2 - 15\log(bL - \log 2)$, for $b = 1/8, \kappa_1$, are increasing and positive. Their powers with exponent δ are therefore increasing. The endpoint primordial inequalities thus guarantee $q_1 \geq 37$ for every large-exceptional branch, irrespective of changes of conductor. No assertion of monotonicity of the conductor itself is made.

Both exceptional alternatives are covered by the common $\mathcal{L}$ and $36/35$ envelopes; no small-case coefficient is used to bound a later large case. Small primes dividing N can change arbitrarily, but $\log Q \leq \bar{q}$ and the local product comparisons hold separately for each N . Hence neither changes of exceptional conductor nor primordial transition points require an unexamined continuity assertion.

10.2 A logarithmic derivative bound for the zero terms

Write $t = bL - \log 2$ and $w = t - 15\log t$, for $b = 1/8, \kappa_1$. For every $c \geq 24.9$,

$$1 < \frac{w'}{w} < 1.01. \quad (49)$$

Indeed

$$w' - w = \log 2 + 15\log t - 15 - \frac{15\log 2}{t} > 0,$$

and, since $w > t/2$,

$$0 < \frac{w'}{w} - 1 < \frac{32\log t}{t} < 0.01.$$

The last bound is checked at the smaller starting t ; $\log t/t$ decreases for $t > e$. For $w = L - 5c$ the same inequality follows from $w'/w = (L - 5)/(L - 5c)$; its excess over one is decreasing here.

In particular the gap needed for the rectangle cutoff, $L/2 - 10c - 10\log w_3$, has derivative $L/2 - 10 - 10w'_3/w_3 > L/2 - 20.1 > 0$. Its certified positive endpoint proves $D_0 < D_R$ throughout. The lower bound $\log D_R \geq L/2 - 10c$ also proves $D_R > 10^9$ throughout. The small-conductor support gaps $\kappa_1 L - 10c$ and $\kappa_1 L - 10\log w_q$ have positive derivatives, so all removed exceptional primes remain below z_* .

The ratio of ν to $1/(2 \cdot 2.0452 \cdot 10c)$ is a constant times $e^{-\delta c/2}/c$, hence decreases. It is below one at $c = 24.9$. All small-exceptional thresholds are at most L^δ , so the common ν is admissible in every stage. Moreover, for each of the three choices of w above,

$$\frac{d}{dc} \log(\nu w) = \frac{w'}{w} - \frac{\delta}{2} - \frac{2}{c} > 1 - \frac{29}{50} - \frac{20}{249} > \frac{1}{3}. \quad (50)$$

The certificate gives $\nu w > 100$ at the smallest starting scale. It follows that $\nu w > 100$ on the entire half-line.

These inequalities also justify every maximum over prime-counting endpoints in (19): for fixed N and $\nu(N)$, differentiate with respect to $x = \log y \geq w$. The derivatives of $x^4 e^{-\nu x}$, $x^{14} e^{-x/2}$ and $x^{14} e^{-2x/3}$ have respective signs of $4 - \nu x$, $14 - x/2$, and $14 - 2x/3$, all negative. The first term x^{-4} decreases too. This proves the endpoint majorant for all intermediate y , not just the endpoint Y .

For variation in N , the logarithmic derivative of $Lw^4 e^{-\nu w}/(1 - \nu)$ is at most $1 + 4(1.01) - \nu w/3 < 0$; the derivative of $(1 - \nu)^{-1}$ only helps. The corresponding derivatives for $Lw^{14} e^{-w/2}$ and $Lw^{14} e^{-2w/3}$ are negative by (49). The derivative of L/w^4 is negative. The multiplier in the second line of (19) decreases, and $L/\log(N^b/2) = 1/(b - \log 2/L)$ decreases. Therefore $L\nu(N^b/2)$, and hence $v(N^b/2)$, decrease. In particular the certified inequality $v(X_q) < 40/L$ holds forever.

10.3 Rectangular and ordinary AP errors

Every term of m_N in (22) decreases: the first by the preceding paragraph; the second because $t^{16}e^{-t}/\log t$ has negative derivative for $t > 16$; the third because $d\log(L^5/w_3^{10})/dc = 5 - 10w'_3/w_3 < -5$; the fourth because $5 - \alpha L < 0$ for $\alpha = 0.199, 1/16$; and the last because $\log X_3$ increases. After summing the ceiling bound, the right side of (24) is exactly a constant times $m_N(A + B/L)$ for fixed positive constants A, B . It therefore decreases. This proves the global box assertion despite the growing number of boxes.

For the AP coefficient, use the explicit formula in section A. Let $w = L - 5c$ and $h = w - 5 \log w$. Besides (49), $h'/h > w'/w > 1$. Each nonconstant summand of $L^2b(N)$ decreases. For the power part of $\mathcal{E}$ its largest derivative is bounded by $2 + (13/2)w'/w - 10h'/h < -3/2$. The other power part has derivative $2 - (7/2)w'/w < -3/2$. The zero term $L^2w^2 \log w e^{-\nu w}/(1 - \nu)$ has logarithmic derivative at most $2 + 2(1.01) + 1.01/\log w - \nu w/3 < 0$. For every remaining exponential term, the highest polynomial degree in w is $15/2$ and its smallest exponential decay is $e^{-w/12}$; $2 + (15/2)(1.01) - w'/12 < 0$ bounds them. The nonexponential term $L^2 \log w/w^6$ has derivative $2 + (1/\log w - 6)w'/w < 0$. Terms with w^{-8} have still smaller derivatives, and the rational multiplier defining p in section A decreases. Finally the added $\omega(N)$ term in $L^2b(N)$ has logarithmic derivative

$$\frac{L-5}{2} - L + 6 - 10 \frac{L-5}{L-5c} - \frac{1}{c} < 0.$$

Thus the endpoint check $L^2b(N) < 2.21$ proves the claimed AP bound for all $c \geq 24.9$. Inserting this into (18) leaves a constant times $(1/L + 1/\log 2)^2/L$, which decreases.

10.4 Switched, local, finite and transfer errors

The normalized high-conductor envelope in the structural certificate is a positive polynomial prefactor of logarithmic derivative at most 6, times the sum of

$$\frac{8}{w_q^{10}}, \quad \sqrt{12}K_R e^{-L/12}, \quad \sqrt{12}K_R e^{-L/3}, \quad 24e^{-0.0846L}.$$

Here $K'_R/K_R < 1$. The first product has derivative less than $6 - 10 = -4$. The other three have derivatives bounded respectively by $7 - L/12$, $7 - L/3$, and $6 - 0.0846L$, all negative. This accounts for every dyadic conductor interval and every n block. For the low-conductor term insert $v(X_q) \leq 40/L$ in (40). After normalization its varying factor is $(1 + L) \log w_q/L^3$, with logarithmic derivative at most $1 + 1.01/\log w_q - 3 < -1.9$.

The Perron norm bound follows from $\operatorname{arsinh}(TL) < \log(2TL) + 1 = 4L + c + \log 2 + 1$. At the endpoint this is below $2\pi L$, and the difference has positive derivative since $2\pi L - (4L + 1) > 0$. The normalized tail is bounded by $e^{20}L^{20}e^{-1.5846L}$, whose derivative has sign $20 - 1.5846L < 0$. The integer rough-sieve remainder $e^{20}L^{-9}$ decreases. Hence the entire switched-error allocation remains valid.

The exceptional mass bound B_E has logarithmic derivative $2/(c \log(10c)) - \delta(L-5)/(L-5c) < 0$, and L^{-2} decreases. In (30), d_N and h_0 decrease term by term. In (33), L^{-2} and L^{-9} decrease, while $LN^{-\kappa_1}$ has derivative of sign $1 - \kappa_1 L < 0$.

For completeness, (35) follows without a numerical claim about a sum of astronomical integers. Since $\log 2 > 1/2$, $W < 2L$, $1 + W < 3L$, and $B < 4N^{11/12}$ here. After division by $4\Theta(N) > 2N/L^2$, the five terms of (34) are bounded by

$$64L^4N^{-1/12}, \quad 12L^4N^{-1/3}, \quad 12L^2N^{-2/3}, \quad 486L^7N^{-11/12}, \quad 36450L^8N^{-1/12}.$$

Their sum is at most $37024L^8N^{-1/12} < e^{30}L^{12}N^{-1/12}$. Its logarithmic derivative is $12 - L/12 < 0$, and its endpoint logarithm is below $\log 10^{-6}$.

This proves every analytic propagation used in (48). The main integrals contain only fixed rational parameters, not varying endpoints in c . All remaining quantities are bounded by the decreasing envelopes just proved. Thus ceiling jumps, changes of source primes, and changes of exceptional conductor are covered at their two sides without assuming that the original discrete expressions are continuous or monotone.

A The AP coefficient used in the certificate

This appendix specifies the BJS formulas used in (7); it also explains why no maximum or hidden choice of threshold is delegated to an uncertified computation. Fix N and put $w = L - 5c$, $\nu = \nu(N)$. For $x \geq w$ define

$$\mathcal{E}_0(x) = \frac{4x^{9/2}}{(x - 5 \log x)^{10}} + \frac{4}{x^{11/2}} + \frac{18e^{-x/12}}{\sqrt{x}} + \frac{5}{2}e^{-x/6}x^{11/2}.$$

The upper coefficient obtained from BJS's bound for the sum of $|E_\psi(y; d, N)|$, with $x = \log y \in [w, L]$, is

$$\begin{aligned} p_2 &= w^2 \left[1.1(10 \log w) \left(\frac{3.2 \cdot 10^{-8}}{w^8} + \frac{e^{-\nu w}}{1 - \nu} \right) + 27\mathcal{E}_0(w) + \frac{e^{-w/2}}{2(\log 2)w^8} + 0.4w^3e^{-w} \right], \\ p_1 &= p_2 + \frac{0.67 + 2e^{-w/6}}{w^8}, \\ p &= p_1 \left(1 + \frac{1}{L^2w^3} + \frac{1}{(1 - 4/w)L} \right) + \frac{2.2}{L^2}, \\ b(N) &= p + \frac{0.9e^{w/2-L}L^4}{w^{10}c}. \end{aligned}$$

These are BJS's p_2, p_1, p, c_4 bounds with a weaker common zero bound. They apply also to the nonexceptional and fixed full-conductor relative sums, since their zero gap is larger.

To justify using the displayed endpoint for p_2 , at fixed N replace $\log Q_1(x_1(N))$ by $10 \log x$ and multiply the BJS bound by x^2/e^x . Every resulting term decreases for $x \geq w$: for the zero term its logarithmic derivative is $2/x + 1/(x \log x) - \nu < 0$; for the leading $\mathcal{E}_0$ power term it is at most $(13/2)/x - 10/x < 0$, since $(1 - 5/x)/(x - 5 \log x) > 1/x$. The remaining powers decrease, and for the exponentials the largest degree is $15/2$ while the smallest decay rate is $1/12$. All threshold inequalities hold already at $w(N_0)$. This proves the maximum over the finite interval $[x_1(N), N]$; extending it to $x \geq w$ only enlarges the allowable majorant.

The passage $\psi \rightarrow \theta$ gives the displayed p_1 ; partial summation from $x_1(N)$ to N and Brun–Titchmarsh at the initial point give the displayed p , including the $2.2/L^2$ term. Finally removing primes dividing N costs at most $1.3841L/c$ per modulus. There are fewer than $0.65H_N$ square-free moduli by BJS's square-free counting bound. Since $0.65 \cdot 1.3841 < 0.9$, their normalized cost is exactly bounded by the last term of $b(N)$. These steps use the proofs of BJS Lemmas 31, 34, 35, and 43, on their stated square-free, coprime modulus ranges. They do not extend the relative result to an arbitrary additional tag.

B Interval integration and reproducibility

The numerical environment is Python 3.12 with `python-flint` 0.9.0, using Arb/ACB at 192-bit precision and one thread. All input constants are integers, exact rational quotients or decimal strings; no binary floating-point approximations enter the certificates. The rational identities are checked with exact rational arithmetic. Every adaptive integral must return a finite real enclosure with at least 90 bits of relative accuracy; an undecided or false comparison aborts

execution. The integration limit is 200000 function evaluations and depth 60, with no fallback to point sampling.

Here are the analytic reductions used to avoid unverifiable nested quadrature. For $3 \leq s \leq 5$,

$$F(s) = \frac{2e^\gamma}{s} \left[1 + \log(s-2) \log(s-1) + \text{Li}_2(2-s) + \frac{\pi^2}{12} \right].$$

Differentiate the bracket to obtain $\log(s-2)/(s-1)$ and use its value 1 at $s=3$; this proves the formula from the delay equation. For $4 \leq s \leq 6$ and $5 \leq s \leq 7$, respectively,

$$f(s) = \frac{2e^\gamma \log 3 + \int_3^{s-1} F(u) du}{s},$$

$$F(s) = \frac{1}{s} \left[5F(5) + 2e^\gamma \log 3 \log \frac{s-1}{4} + \int_3^{s-2} F(u) \log \frac{s-1}{u+1} du \right].$$

The second identity follows by inserting the first in $sF(s) = 5F(5) + \int_5^s f(v-1) dv$ and reversing the finite integrals. These identities supply every function value beyond the printed elementary $s \leq 4$ range.

For L_4 , reflect the symmetric triangle to half of its square and then set $v = t + u$. The one-dimensional weight is

$$\begin{cases} v^{-1} \log((v - \kappa_1)/\kappa_1), & 2\kappa_1 \leq v \leq \kappa_1 + \kappa_2, \\ v^{-1} \log(\kappa_2/(v - \kappa_2)), & \kappa_1 + \kappa_2 \leq v \leq 2\kappa_2. \end{cases}$$

For L_5 the corresponding weight is

$$\begin{cases} v^{-1} \log((v - \kappa_2)(v - \kappa_1)/(\kappa_1 \kappa_2)), & \kappa_1 + \kappa_2 \leq v \leq 2\kappa_2, \\ v^{-1} \log(\kappa_2(v - \kappa_1)/(\kappa_1(v - \kappa_2))), & 2\kappa_2 \leq v \leq \lambda. \end{cases}$$

The formulas follow by integrating $1/(t(v-t))$ over the relevant line section, whose antiderivative is $v^{-1} \log(t/(v-t))$. The A_2, A_3 integrals are split where their arguments equal 3 and 2; the latter split is new because the retuned upper endpoints enter the branch $1 < s < 2$. The exceptional L_5 integral is split at exactly $v = \theta_- - 3\kappa_1$ and $v = \theta_- - 2\kappa_1$, not at the unshifted values. The latter is the jump point of $\mathcal{L}$. The other integrals are split wherever their delay-function formula changes. Thus no nonsmooth formula is passed across an unmarked integration boundary.

The sole forward entry point is

```
python c249_verify.py --check-tex.
```

It executes both component certificates afresh, reconstructs the signed rows of (47), checks every printed interval against the new ball enclosure, sums the seven budgets, and writes `c249_final_ledger.json`. That file records directions, Wu weights, normalization, complete balls, engine precision and SHA-256 hashes of the scripts, this manuscript, and the exact two external source files used. The component JSON files are reproducible subordinate outputs, not alternative ledgers. Previous 23.36 and diagnostic floating-point ledgers are not inputs to this proof.

The numerical run is not a formal proof checker for the analytic lemmas. Their proofs and their hypotheses are the body of this manuscript. The role of the run is solely to enclose the finite computations and certify the explicit starting inequalities that the half-line argument propagates.

References

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